

TrES-2b

R-2441

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-2_120913 · created 2026-08-22T22:49:15+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456183.65788 +/- 0.00018 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.148 +/- 0.039
Transit depth (Rp/Rs)^2	2.21 +/- 1.17 [%]
Orbital Inclination (inc)	82.8 +/- 2.64
Airmass coefficient 1 (a1)	1.068 +/- 0.011
Airmass coefficient 2 (a2)	-0.061 +/- 0.049
Scatter in the residuals of the lightcurve fit is	1.07 %
Best Comparison Star	#2 - [130, 129]
Optimal Aperture	3.47
Optimal Annulus	9.19
Transit Duration (day)	0.049 +/- 0.038

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.148 ± 0.039 vs 0.1254 (deviation 0.41σ)
Duration fit vs published	0.049 d vs 0.0746 d (deviation 0.47σ)
Mid-time O–C	-0.4 min
Depth SNR	2.7
Residual scatter	1.07 %

Light curve

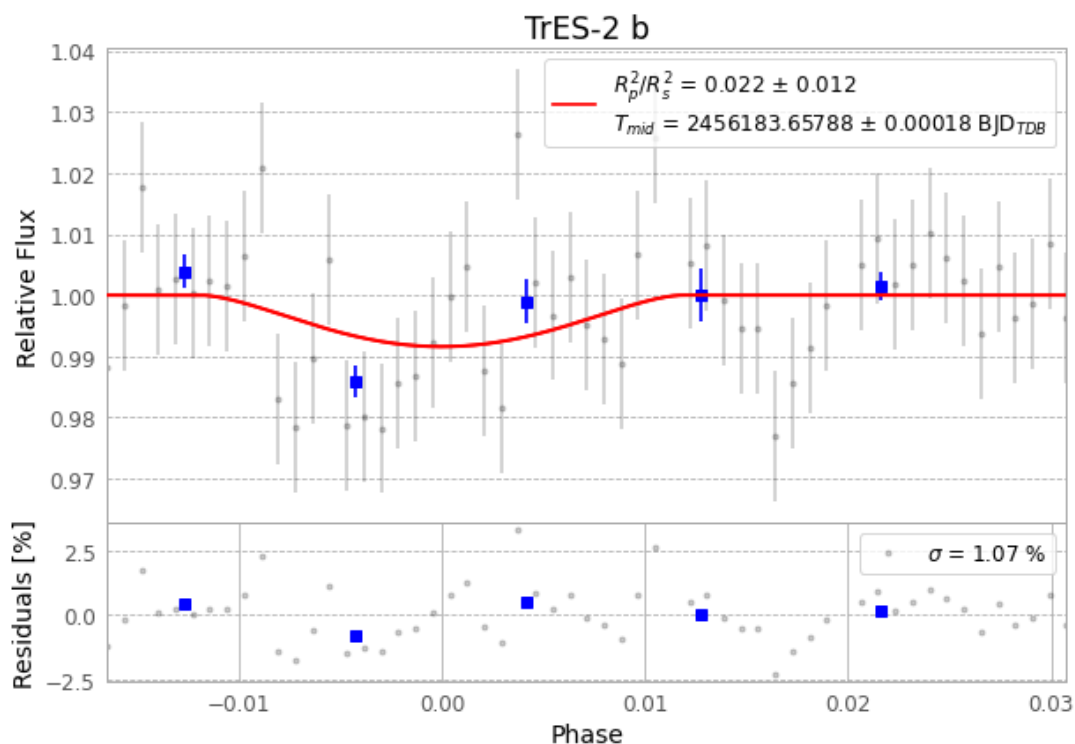


Figure 1 — FinalLightCurve_TrES-2 b_2012-09-12.png (SHA-256 9e7e904de1f14ff4...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [403, 258], 2 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 382ffb31768725d4...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

57 FITS frames (37 MB), TRES-2120913024512.FITS → TRES-2120913053312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-2-120913.log (6a5b58da7d54...), FinalLightCurve_TrES-2 b_2012-09-12.png (9e7e904de1f1...), FinalLightCurve_TrES-2 b_2012-09-12.csv (b53ed15cbc8a...)

Compute resources

Wall time 184.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-22 22:49Z.